# Lecture L17 Scribe CSE400

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Generate a lecture scribe for Lecture L17 of CSE400 – Fundamentals of Probability in Computing.

The scribe must function as formal exam-preparation reference material, allowing a student to revise the lecture accurately using only this document.

Use only the provided context (lecture slides and any attached textbook excerpts). Do not introduce new explanations, examples, assumptions, or results that are not present in the provided materials.

While writing the lecture scribe, reason step-by-step through the lecture content and include the reasoning in the scribe so that the logical development of the lecture is clear for revision. All reasoning must be derived strictly from the provided context.

Follow these requirements:

Maintain a formal academic tone suitable for exam preparation.

Follow the structure and logical order of the lecture as presented in the provided materials.

Clearly present:

Definitions and notation

Assumptions and conditions

Statements of results or properties

Proofs or derivations (step-by-step when present in the lecture)

Worked examples with intermediate steps exactly as shown in the lecture

Ensure that the logical reasoning behind definitions, derivations, and examples is explicitly explained step-by-step, but only using information present in the provided context.

Organize the scribe using clear sections and headings reflecting the lecture structure.

Preserve mathematical notation and formulas accurately.

Do not invent additional explanations, examples, proofs, or interpretations that are not present in the lecture material.

The final output should be a complete, well-structured lecture scribe that faithfully reconstructs the lecture content and reasoning for reliable exam revision.

Below is a **faithful lecture scribe for Lecture L17** based strictly on the provided slides. The structure, definitions, derivations, and examples follow the lecture order exactly and only include information present in the slides. The lecture slides are taken from the uploaded document L15_L17.

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# Lecture Scribe

# CSE400 – Fundamentals of Probability in Computing

### Lecture L17: Joint Random Variables and Joint Distributions

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## 1. Introduction: Why Study Joint Random Variables?

In many real-world situations, we are interested in **two or more random variables simultaneously** rather than analyzing a single random variable. Studying such variables allows us to understand relationships between different stochastic quantities. L15_L17

### Real-World Examples

Examples where two random variables must be studied together include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members L15_L17

The purpose of studying these variables jointly is to determine **how they are related**.

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## 1.1 Example Applications

### Smart Transportation System

Two random variables are defined:

- $X$: Vehicle Speed
- $Y$: Traffic Density

Key question:

> Can vehicle speed be high when traffic density is also high?

The answer may vary depending on conditions, indicating that $X$ **and** $Y$ **are correlated random variables**.

Applications include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

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### Wireless Network Performance

Two random variables are considered:

- $X$: Signal-to-Noise Ratio (SNR)
- $Y$: Data Throughput

Observation:

Throughput depends on SNR but may also depend on congestion.

Question:

> If SNR is high, will throughput always be high?

Answer: Not necessarily.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

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### Additional Examples

Other scenarios involving joint random variables include:

- Weather prediction
  - $X$: Rainfall
  - $Y$: Temperature
- Drone delivery
  - $X$: Wind speed
  - $Y$: Delivery time
- Rolling two dice
  - $X$: value of one die
  - $Y$: value of another die

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## 1.2 Purpose of Joint Distributions

When random variables are considered together:

- They have a **joint distribution**.
- This distribution allows computation of probabilities involving both variables.
- It helps analyze the **relationship between the variables**.

Special cases:

- If variables are independent, analysis is simpler.
- If variables are dependent, measures such as **covariance and correlation** are used.

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## 2. Discrete Case: Joint Probability Mass Function (Joint PMF)

## 2.1 Setup

Suppose two discrete random variables:

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

The ordered pair $(X, Y)$ takes values from the **product set**

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

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## 2.2 Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Interpretation:

This represents the probability that **both events occur simultaneously**, meaning:

- $X$ takes value $x_i$
- $Y$ takes value $y_j$

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## 2.3 Joint PMF Table Representation

The joint PMF is commonly represented in a **table format**.

Rows correspond to values of $X$ and columns correspond to values of $Y$.

Example table structure:

| X \ Y | $y_1$ | $y_2$ | ... | $y_j$ | ... | $y_m$ |
|---|---|---|---|---|---|---|
| $x_1$ | $p(x_1,y_1)$ | $p(x_1,y_2)$ | ... | $p(x_1,y_j)$ | ... | $p(x_1,y_m)$ |

| X \ Y | $y_1$ | $y_2$ | ... | $y_j$ | ... | $y_m$ |
|---|---|---|---|---|---|---|
| $x_2$ | $p(x_2,y_1)$ | $p(x_2,y_2)$ | ... | $p(x_2,y_j)$ | ... | $p(x_2,y_m)$ |
| ... | ... | ... | ... | ... | ... | ... |

### Key Properties

1. Probabilities are bounded:

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability equals one:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

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## 3. Example: Rolling Two Dice

### Experiment

Roll two fair dice.

Define random variables:

- $X$: value on the first die
- $T$: total sum of both dice

Each ordered outcome $(i, j)$ occurs with probability:

$$\frac{1}{36}$$

Therefore, we construct the **joint probability table for** $(X, T)$. L15_L17

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## Observations from the Joint Table

From the table shown in the lecture slides:

- Entries are either **0 or** $1/36$.
- Variables $X$ and $T$ are **not independent**.

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## 4. Continuous Case: Joint Probability Density Function (Joint PDF)

The continuous case is analogous to the discrete case.

### Correspondence Between Cases

| Discrete | Continuous |
|---|---|
| Discrete values | Continuous intervals |
| Joint PMF | Joint PDF |
| Sums | Double integrals |

If:

$$X \in [a, b], \quad Y \in [c, d]$$

then

$$(X, Y) \in [a, b] \times [c, d]$$

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## 4.1 Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** of $X$ and $Y$ if

$$P((X, Y) \text{ lies in small rectangle}) \approx f(x, y)\, dx\, dy$$

Thus, the probability over a small region equals:

$$f(x, y)\, dx\, dy$$

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## 4.2 Key Properties

1. Non-negativity

$$f(x, y) \geq 0$$

2. Normalization condition

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Important note:

A density value $f(x, y)$ **may be greater than 1**, since it represents a **density**, not a probability.

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## 5. Worked Example

Given joint PDF:

$$f(x, y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

The pair $(X, Y)$ lies in the **unit square**

$$[0, 1]^2$$

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## Task 1: Verify Valid Joint PDF

Compute the normalization integral.

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Step 1:

$$\int_0^1 \left( \int_0^1 4xy \, dx \right) dy$$

Step 2:

$$\int_0^1 2y \, dy$$

Step 3:

$$[y^2]_0^1 = 1$$

Thus:

- $f(x, y) \geq 0$
- total probability = 1

Therefore, $f(x, y)$ is a **valid joint PDF**.

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## Task 2: Compute Probability of Event

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Compute:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

Step 1:

$$\int_0^{0.5} [2xy^2]_{0.5}^1 dx$$

Step 2:

$$\int_0^{0.5} 2x(1-0.25)dx$$

Step 3:

$$\int_0^{0.5} \frac{3}{2}x\,dx$$

Final result:

$$P(A) = \frac{3}{16}$$

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## 6. Joint Cumulative Distribution Function (Joint CDF)

Suppose $X$ and $Y$ are jointly distributed random variables.

Notation:

$$X \leq x, \quad Y \leq y$$

represents the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

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## Definition

The **joint cumulative distribution function** is defined as

$$F(x,y) = P(X \leq x, Y \leq y)$$

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## 7. Joint CDF for Continuous Variables

If $X$ and $Y$ are continuous random variables with joint PDF $f(x,y)$, then

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv$$

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## Recovering the Joint PDF

The joint PDF can be obtained using **partial derivatives**:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

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## 8. Joint CDF for Discrete Variables

If $X$ and $Y$ are discrete random variables with joint PMF $p(x_i, y_j)$, then

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation:

To compute the joint CDF:

- Add probabilities from the joint PMF table
- Include rows where $x_i \leq x$
- Include columns where $y_j \leq y$ L15_L17

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## 9. Properties of the Joint CDF

The joint CDF $F(x, y)$ satisfies the following properties:

### 1. Monotonicity

$$F(x, y) \text{ is non-decreasing}$$

If $x$ or $y$ increases, $F(x, y)$ remains constant or increases.

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### 2. Lower Boundary Condition

$$F(x, y) = 0$$

at the lower-left corner of the joint range.

If the lower bound is $(-\infty, -\infty)$, then

$$\lim_{(x,y) \to (-\infty, -\infty)} F(x, y) = 0$$

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### 3. Upper Boundary Condition

$$F(x, y) = 1$$

at the upper-right corner of the joint range.

If the upper bound is $(\infty, \infty)$, then

$$\lim_{(x,y)\to(\infty,\infty)} F(x,y) = 1$$

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## 10. Summary of Lecture

This lecture introduced the concept of **joint random variables and joint distributions**.

Main ideas covered:

1. Motivation for studying joint random variables
2. Joint PMF for discrete variables
3. Joint PMF table representation
4. Example using two dice
5. Joint PDF for continuous variables
6. Example computation with a joint PDF
7. Definition of the joint CDF
8. Continuous and discrete forms of joint CDF
9. Properties of the joint CDF

These concepts form the foundation for analyzing **relationships between multiple random variables in probabilistic systems**.

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If you want, I can also convert this **directly into the exact LaTeX lecture scribe format your professor expects (same as the previous scribe assignments)** so you can **submit it immediately without editing.**

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# Lecture Scribe

# CSE400 – Fundamentals of Probability in Computing

### Lecture L17: Joint Random Variables and Joint Distributions

This lecture introduces the concept of **joint random variables and joint distributions**. The goal is to analyze situations where **two random variables are considered simultaneously** and to study the probability structure that governs their joint behavior. The following scribe reconstructs the lecture structure and reasoning based strictly on the provided slides.

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## 1. Motivation: Why Study Joint Random Variables?

In scientific analysis and real-world applications, it is often necessary to study **two or more random variables at the same time**. Studying them jointly allows the analysis of relationships between these variables.
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Examples of such situations include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples illustrate situations where the outcome of one variable may be related to another variable.
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# 1.1 Illustrative Applications

## Smart Transportation System

Two random variables are defined:

- $X$ = Vehicle Speed
- $Y$ = Traffic Density

A key question arises:

**Can vehicle speed be high when traffic density is also high?**

The possibility of such interactions indicates that $X$ and $Y$ may be **correlated random variables**.

Applications of such analysis include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

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## Wireless Network Performance

Two random variables are considered:

- $X$ = Signal-to-Noise Ratio (SNR)
- $Y$ = Data Throughput

Throughput depends on SNR but may also depend on other factors such as network congestion.

Key question:

**If SNR is high, will throughput always be high?**

The answer is not necessarily yes, since throughput may also be influenced by congestion and interference.
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Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

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### Additional Examples

Other systems involving joint random variables include:

- Weather prediction
  - $X$ = Rainfall
  - $Y$ = Temperature
- Drone delivery
  - $X$ = Wind speed
  - $Y$ = Delivery time
- Rolling two dice

These examples show that many real-world systems involve **multiple random variables simultaneously**.
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## 1.2 Purpose of Joint Distributions

When random variables are studied together:

- They have a **joint distribution**.
- The joint distribution allows computation of probabilities involving both variables.
- It helps us understand the **relationship between variables**.

Two cases arise:

- When variables are **independent**, analysis is simpler.
- When variables are **dependent**, measures such as **covariance and correlation** describe the relationship between them.
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# 2. Discrete Case: Joint Probability Mass Function (Joint PMF)

## 2.1 Setup

Assume $X$ and $Y$ are discrete random variables.

Let

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

The ordered pair $(X, Y)$ takes values in the **product set**

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Thus, every outcome corresponds to a combination of values of $X$ and $Y$. L15_L17

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## 2.2 Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Interpretation:

This function gives the probability that **both random variables simultaneously take the specified values**. L15_L17

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## 3. Joint PMF Table Representation

The joint PMF can be organized into a table where:

- Rows correspond to values of $X$
- Columns correspond to values of $Y$

Each cell contains the value:

$$p(x_i, y_j)$$

### Key Properties

1. Probability bounds

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability equals 1

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

This ensures that the probabilities across all possible joint outcomes sum to one. L15_L17

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## 4. Example: Rolling Two Dice

### Experiment

Roll two fair dice.

Define random variables:

- $X$ = value on the first die
- $T$ = total (sum) of both dice

Each pair of outcomes $(i, j)$ occurs with probability

$$\frac{1}{36}$$

Using this information, we construct the **joint probability table for** $(X, T)$. L15_L17

The table shown in the lecture slides lists all combinations of $X$ and $T$.

### Observations

From the joint probability table:

- Entries are either $0$ or $1/36$
- $X$ and $T$ are **not independent** random variables

This occurs because the value of $T$ depends on the value of $X$. L15_L17

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## 5. Continuous Case: Joint Probability Density Function (Joint PDF)

The continuous case follows an analogy with the discrete case.

### Correspondence Between Discrete and Continuous Cases

| Discrete Case | Continuous Case |
|---|---|
| Discrete values | Continuous intervals |
| Joint PMF | Joint PDF |
| Sums | Double integrals |

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair $(X, Y)$ lies in the **product region**

$$[a, b] \times [c, d]$$

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## 5.1 Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** of $X$ and $Y$ if

$$P((X, Y) \text{ lies in a small rectangle of area } dx\,dy) \approx f(x, y)\,dx\,dy$$

Thus the probability associated with a small rectangular region is approximated by the density multiplied by its area.

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## 5.2 Key Properties of the Joint PDF

1. Non-negativity

$$f(x, y) \geq 0$$

2. Normalization condition

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

Important remark:

The value of $f(x, y)$ **may exceed 1**, because it represents a **density**, not a probability.

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## 6. Worked Example

Given

$$f(x, y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Thus, $(X, Y)$ takes values in the **unit square**

$$[0, 1]^2$$

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## Task 1: Verify Validity of the Joint PDF

To verify validity, check the normalization condition.

$$\int_0^1 \int_0^1 4xy\,dx\,dy$$

Step 1:

$$\int_0^1 \left( \int_0^1 4xy \, dx \right) dy$$

Step 2:

$$\int_0^1 2y \, dy$$

Step 3:

$$[y^2]_0^1 = 1$$

Since the integral equals 1 and $f(x, y) \geq 0$, the function is a **valid joint PDF**.

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## Task 2: Compute Probability of Event

Event:

$$A = \{X < 0.5, \ Y > 0.5\}$$

Compute

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

Step-by-step computation:

$$P(A) = \int_0^{0.5} [2xy^2]_{0.5}^1 dx$$

$$= \int_0^{0.5} 2x(1 - 0.25) dx$$

$$= \int_0^{0.5} \frac{3}{2} x \, dx$$

$$= \frac{3}{16}$$

Thus,

$$P(A) = \frac{3}{16}$$

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## 7. Joint Cumulative Distribution Function (Joint CDF)

Suppose $X$ and $Y$ are jointly distributed random variables.

Notation:

$$X \leq x, \quad Y \leq y$$

represents the event

$$\{X \leq x \text{ and } Y \leq y\}$$

### Definition

The **joint cumulative distribution function** is

$$F(x, y) = P(X \leq x, Y \leq y)$$

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## 8. Joint CDF – Continuous Case

If $X$ and $Y$ are continuous random variables with joint density $f(x, y)$, then

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Thus the joint CDF is obtained by integrating the joint density over the region defined by $u \leq x$ and $v \leq y$.

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## Recovering the Joint PDF

Since there are two variables, the density can be recovered using partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

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## 9. Joint CDF – Discrete Case

If $X$ and $Y$ are discrete random variables with joint PMF $p(x_i, y_j)$, then

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

### Interpretation

To compute $F(x, y)$:

1. Add probabilities in the joint PMF table.
2. Include rows where $x_i \leq x$.
3. Include columns where $y_j \leq y$.

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# 10. Properties of the Joint CDF

The joint CDF satisfies several properties.

### Property 1: Non-decreasing

$$F(x, y)$$

is **non-decreasing**.
If $x$ or $y$ increases, the value of $F(x, y)$ remains constant or increases.
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### Property 2: Lower-left boundary

$$F(x, y) = 0$$

at the lower-left corner of the joint range.

If the lower bound is $(-\infty, -\infty)$,

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

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### Property 3: Upper-right boundary

$$F(x, y) = 1$$

at the upper-right corner of the joint range.

If the upper bound is $(\infty, \infty)$,

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

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# 11. Summary

This lecture established the framework for studying **joint random variables**.

Main topics covered:

1. Motivation for joint random variables
2. Joint PMF for discrete variables
3. Joint probability table representation
4. Example involving two dice
5. Joint PDF for continuous variables
6. Verification and probability calculation example

7. Definition of the joint CDF
8. Continuous and discrete formulations of the joint CDF
9. Fundamental properties of the joint CDF

These concepts form the basis for analyzing systems involving **multiple random variables simultaneously**.
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# Lecture Scribe

# CSE400 – Fundamentals of Probability in Computing

### Lecture L17: Joint Random Variables and Joint Distributions

This lecture introduces **joint random variables**, their **joint distributions**, and the tools used to analyze probabilities involving two variables simultaneously. The presentation follows the exact order and reasoning provided in the lecture slides. L15_L17

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## 1. Motivation: Why Study Joint Random Variables?

In many scientific and real-world scenarios, it is necessary to study **two or more random variables at the same time**. Studying them jointly enables us to compute probabilities of events involving both variables and understand the **relationship between them**. L15_L17

Examples where two variables are naturally studied together include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members L15_L17

These situations motivate the need for **joint probability models**.

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## 1.1 Example Applications

### Example 1: Smart Transportation

Consider a smart traffic system with two random variables:

- $X$ = Vehicle Speed

- $Y$ = Traffic Density

Question posed in the lecture:

**Can speed be high when traffic density is also high?**

The possible relationship between these variables implies that $X$ **and** $Y$ **may be correlated random variables**.

Applications mentioned in the lecture include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

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## Example 2: Wireless Network Performance

Two random variables are considered:

- $X$ = Signal-to-Noise Ratio (SNR)
- $Y$ = Data Throughput

Reasoning:

Throughput depends on SNR, but both variables vary due to factors such as fading, interference, and congestion.

Question posed:

**If SNR is high, will throughput always be high?**

The lecture notes that this may not always occur because network congestion can influence throughput.

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Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

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## Additional Examples

Other examples presented include:

- Weather prediction
  - $X$ = Rainfall
  - $Y$ = Temperature
- Drone delivery
  - $X$ = Wind Speed

- $Y$ = Delivery Time
- Rolling two dice, where events such as $X + Y = 7$ and $X > Y$ require a **joint PMF**.

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## 1.2 Purpose of Joint Distributions

When two random variables are studied together:

- They have a **joint distribution**.
- This distribution allows computation of probabilities involving both variables.
- It helps understand the **relationship between variables**.

The lecture notes two cases:

- If variables are **independent**, analysis is simpler.
- If variables are **dependent**, measures such as **covariance and correlation** describe the dependence between them.

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# 2. Discrete Case – Joint Probability Mass Function (Joint PMF)

## 2.1 Setup

Suppose $X$ and $Y$ are **discrete random variables**.

Let

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

The ordered pair $(X, Y)$ takes values in the **product set**

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Thus every outcome of the joint system corresponds to a pair of values from the domains of $X$ and $Y$.

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## 2.2 Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Interpretation:

This represents the probability that **both random variables simultaneously take the specified values**.

L15_L17

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## 3. Joint PMF Table Representation

The joint PMF is commonly organized in a **table format**.

- Rows correspond to values of $X$
- Columns correspond to values of $Y$
- Each cell contains $p(x_i, y_j)$

### Key Properties

1. Probability bounds:

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability equals one:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

This ensures that the sum of probabilities over all possible joint outcomes equals one. L15_L17

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## 4. Example: Rolling Two Dice

### Experiment

Roll two fair dice.

Define:

- $X$ = value on the first die
- $T$ = total (sum) of both dice

Each ordered outcome $(i, j)$ has probability

$$\frac{1}{36}$$

Using this, a **joint probability table for** $(X, T)$ is constructed in the lecture slides. L15_L17

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## Observations from the Table

The lecture notes the following observations:

- Entries in the table are either $0$ or $1/36$.
- The variables $X$ and $T$ are **not independent**.

This occurs because the value of $T$ depends directly on the value of $X$. L15_L17

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# 5. Continuous Case – Joint Probability Density Function (Joint PDF)

The continuous case is presented as an **analogy to the discrete case**.

### Correspondence

| Discrete Case | Continuous Case |
|---|---|
| Discrete values | Continuous intervals |
| Joint PMF | Joint PDF |
| Sums | Double integrals |

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair $(X, Y)$ lies in the **product region**

$$[a, b] \times [c, d]$$

L15_L17

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## 5.1 Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** of $X$ and $Y$ if

$$P((X, Y) \text{ lies in a small rectangle of area } dx\, dy) \approx f(x, y)\, dx\, dy$$

Thus the probability of a small region is approximated by the density multiplied by its area. L15_L17

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## 5.2 Key Properties

1. Non-negativity

$$f(x, y) \geq 0$$

2. Normalization condition

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Remark from the lecture:

The value $f(x, y)$ **may exceed 1**, since it is a density rather than a probability. L15_L17

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## 6. Worked Example

Given joint density:

$$f(x, y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Thus $(X, Y)$ lies in the **unit square**

$$[0, 1]^2$$

L15_L17

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## Task 1: Verify Valid Joint PDF

Check normalization:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Step 1

$$\int_0^1 \left( \int_0^1 4xy \, dx \right) dy$$

Step 2

$$\int_0^1 2y \, dy$$

Step 3

$$[y^2]_0^1 = 1$$

Conclusion:

- $f(x, y) \geq 0$
- Total probability = 1

Therefore $f(x, y)$ is a **valid joint PDF**.

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## Task 2: Compute Probability of Event

Event:

$$A = \{X < 0.5, \ Y > 0.5\}$$

Compute

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

Step-by-step computation:

$$P(A) = \int_0^{0.5} [2xy^2]_{0.5}^{1} dx$$

$$= \int_0^{0.5} 2x(1 - 0.25) dx$$

$$= \int_0^{0.5} \frac{3}{2} x \, dx$$

$$P(A) = \frac{3}{16}$$

L15_L17

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## 7. Joint Cumulative Distribution Function (Joint CDF)

Suppose $X$ and $Y$ are jointly distributed random variables.

Notation:

$$X \leq x, \quad Y \leq y$$

represents the event

$$\{X \leq x \text{ and } Y \leq y\}$$

### Definition

The **joint cumulative distribution function** is

$$F(x, y) = P(X \leq x, Y \leq y)$$

L15_L17

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## 8. Joint CDF – Continuous Case

If $X$ and $Y$ have joint density $f(x, y)$, then

$$F(x, y) = \int_c^y \int_a^x f(u, v) \, du \, dv$$

Thus the joint CDF is obtained by integrating the joint density over the appropriate region.

L15_L17

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## Recovering the Joint PDF

Since two variables are involved, the density can be recovered using partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

L15_L17

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## 9. Joint CDF – Discrete Case

If $X$ and $Y$ are discrete random variables with joint PMF $p(x_i, y_j)$, then

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

### Interpretation

To compute $F(x, y)$:

- Add probabilities in the joint PMF table
- Include rows with $x_i \leq x$
- Include columns with $y_j \leq y$ L15_L17

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## 10. Properties of the Joint CDF

The joint CDF $F(x, y)$ satisfies the following properties.

### Property 1: Non-decreasing

$$F(x, y)$$

is **non-decreasing**.

If $x$ or $y$ increases, the value of $F(x, y)$ remains the same or increases. L15_L17

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### Property 2: Lower-left boundary

$$F(x, y) = 0$$

at the lower-left corner of the joint range.

If the lower bound is $(-\infty, -\infty)$,

$$\lim_{(x,y) \to (-\infty, -\infty)} F(x, y) = 0$$

L15_L17

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### Property 3: Upper-right boundary

$$F(x, y) = 1$$

at the upper-right corner of the joint range.

If the upper bound is $(\infty, \infty)$,

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

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## 11. Summary

Lecture L17 introduced the framework for analyzing **two random variables simultaneously**.

Key topics covered:

1. Motivation for joint random variables
2. Joint PMF for discrete variables
3. Joint PMF table representation
4. Example using two dice
5. Joint PDF for continuous variables
6. Validation and probability computation example
7. Definition of the joint CDF
8. Continuous and discrete formulations of the joint CDF
9. Properties of the joint CDF

These concepts form the foundation for further study of **dependence between random variables and multivariate probability models**.

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If you want, I can also give you a **clean LaTeX version of this exact scribe (same format used for CSE400 lecture-scribe submissions)** so you can **submit it directly without editing**.

**Sources**

**L15_L17.pdf**
PDF

Generate a lecture scribe for Lecture L17 of CSE400 – Fundamentals of Probability in Computing.

The scribe must function as formal exam-preparation reference material, allowing a student to revise the lecture accurately using only this document.

Use only the provided context (lecture slides and any attached textbook excerpts). Do not introduce new explanations, examples, assumptions, or results that are not present in the provided materials.

While writing the lecture scribe, reason step-by-step through the lecture content and include the reasoning in the scribe so that the logical development of the lecture is clear for revision. All reasoning must be derived strictly from the provided context.

Follow these requirements:

Maintain a formal academic tone suitable for exam preparation.

Follow the structure and logical order of the lecture as presented in the provided materials.

Clearly present:

Definitions and notation

Assumptions and conditions

Statements of results or properties

Proofs or derivations (step-by-step when present in the lecture)

Worked examples with intermediate steps exactly as shown in the lecture

Ensure that the logical reasoning behind definitions, derivations, and examples is explicitly explained step-by-step, but only using information present in the provided context.

Organize the scribe using clear sections and headings reflecting the lecture structure.

Preserve mathematical notation and formulas accurately.

Do not invent additional explanations, examples, proofs, or interpretations that are not present in the lecture material.

The final output should be a complete, well-structured lecture scribe that faithfully reconstructs the lecture content and reasoning for reliable exam revision.

# Lecture Scribe

# CSE400 – Fundamentals of Probability in Computing

## Lecture L17: Joint Random Variables and Joint Distributions

This lecture introduces the concept of **joint random variables** and the corresponding **joint distributions** used to model the probability behavior of two random variables simultaneously. The lecture explains the motivation for studying such variables, defines the joint probability mass function (PMF) and joint probability density function (PDF), illustrates these ideas through examples, and introduces the **joint cumulative distribution function (CDF)** together with its properties. The exposition below follows the logical order of the lecture slides and reconstructs the reasoning presented during the lecture.

# 1. Motivation: Why Study Joint Random Variables?

In many scientific and engineering applications, we are not interested in a single random variable but rather **two or more random variables simultaneously**. Studying them jointly allows us to compute probabilities involving multiple variables and understand how these variables interact or depend on each other.

L15_L17

The lecture provides several motivating examples where two random variables naturally arise together:

- **Height and weight of giraffes**
- **IQ and birthweight of children**
- **Frequency of exercise and rate of heart disease**
- **Air pollution level and respiratory illness rate**
- **Number of Facebook friends and age of members**

These examples illustrate situations where two measurable quantities are observed simultaneously, and the relationship between them becomes important. The lecture also prompts the question:

> What type of relationship might exist between these variables?

Understanding such relationships requires a framework that allows probabilities to be assigned to **pairs of outcomes**, which motivates the concept of **joint distributions**.

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# 2. Real-World Motivating Examples

The lecture presents several real-world systems where joint random variables arise naturally.

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## 2.1 Smart Transportation System

Consider a smart traffic system involving two random variables:

- $X$: Vehicle Speed
- $Y$: Traffic Density

The lecture raises the question:

> Can vehicle speed be high when traffic density is also high?

The answer may vary depending on traffic conditions. Because the two variables influence each other, they may be **correlated random variables**.

This type of probabilistic analysis has practical applications in **Intelligent Transportation Systems (ITS)**, including:

- Adaptive traffic signal control
- Traffic congestion prediction

- Autonomous vehicle route planning

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## 2.2 Wireless Network Performance

Another example involves wireless communication systems.

Two random variables are defined:

- $X$: Signal-to-Noise Ratio (SNR)
- $Y$: Data Throughput

Throughput depends on the signal quality represented by SNR. However, even when SNR is high, throughput may still be limited due to other factors such as **network congestion**.

The lecture asks:

> If SNR is high, will throughput always be high?

The answer is not necessarily yes. This illustrates that the variables are related but not deterministically linked.

Applications mentioned in the lecture include:

- 5G network scheduling
- Adaptive modulation schemes
- Network planning and performance optimization

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## 2.3 Additional Examples

Several additional scenarios involving joint random variables are mentioned:

1. **Weather Prediction**
   - $X$: Rainfall
   - $Y$: Temperature
2. **Drone Delivery Systems**
   - $X$: Wind Speed
   - $Y$: Delivery Time
3. **Rolling Two Dice**
   - $X$: Value of one die
   - $Y$: Value of another die

For example, events such as

$$X + Y = 7 \quad \text{or} \quad X > Y$$

require a probability model involving both variables simultaneously.

L15_L17

## 3. Purpose of Joint Distributions

When two random variables are analyzed together, they have a **joint distribution**.

The lecture highlights several reasons for studying joint distributions:

1. They allow computation of probabilities involving both variables.
2. They enable analysis of the **relationship between variables**.
3. They help determine whether variables are independent or dependent.

The slides emphasize that:

- Analysis is simplest when variables are **independent**.
- When variables are not independent, measures such as **covariance** and **correlation** are used to describe their dependence.

L15_L17

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## 4. Discrete Case: Joint Probability Mass Function (Joint PMF)

The lecture first considers the case where the random variables are **discrete**.

---

## 4.1 Setup

Suppose $X$ and $Y$ are discrete random variables.

Let the possible values of $X$ be

$$X = \{x_1, x_2, \ldots, x_n\}$$

and the possible values of $Y$ be

$$Y = \{y_1, y_2, \ldots, y_m\}$$

When both variables are considered together, the pair $(X, Y)$ can take values from the **product set**

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Thus each outcome of the experiment corresponds to an ordered pair representing the values of both variables simultaneously.

L15_L17

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## 4.2 Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability that both events occur at the same time:

- the random variable $X$ takes the value $x_i$
- the random variable $Y$ takes the value $y_j$

Hence, the joint PMF describes the probability of each possible **joint outcome**. L15_L17

---

## 5. Joint PMF Table Representation

The joint PMF is often represented using a **table**.

- Rows correspond to values of $X$
- Columns correspond to values of $Y$
- Each cell contains the probability $p(x_i, y_j)$

The lecture notes two important properties of the joint PMF:

### Property 1: Probability bounds

$$0 \leq p(x_i, y_j) \leq 1$$

### Property 2: Total probability equals 1

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

This ensures that the probabilities assigned to all possible joint outcomes sum to one. L15_L17

---

## 6. Example: Rolling Two Dice

The lecture provides an example to illustrate the joint PMF.

---

## 6.1 Experiment

Roll two fair dice.

Define:

- $X$: value on the first die
- $T$: total (sum) of both dice

Each ordered outcome $(i, j)$ occurs with probability

$$\frac{1}{36}$$

since there are 36 equally likely outcomes. L15_L17

---

## 6.2 Joint Probability Table

Using the definition of $X$ and $T$, the lecture constructs the **joint probability table of** $(X, T)$.

The table lists all combinations of values for $X$ and $T$, and each cell contains the probability associated with that pair.

The slide (page showing the table) indicates two observations:

1. Entries in the table are either **0** or $1/36$.
2. The variables $X$ and $T$ are **not independent**.

The lack of independence arises because the value of the sum $T$ depends directly on the value of $X$.

L15_L17

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# 7. Continuous Case: Joint Probability Density Function (Joint PDF)

The lecture next considers the case where the variables are **continuous**.

---

## 7.1 Analogy with the Discrete Case

The continuous case parallels the discrete case in the following way:

| Discrete Case | Continuous Case |
|---|---|
| Discrete values | Continuous intervals |
| Joint PMF | Joint PDF |
| Summation | Double integration |

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair $(X, Y)$ lies in the **product region**

$$[a, b] \times [c, d]$$

L15_L17

---

## 7.2 Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** of $X$ and $Y$ if

$$P((X, Y) \text{ lies in a small rectangle of area } dx\, dy) \approx f(x, y)\, dx\, dy$$

This means that the probability of a small region around $(x, y)$ is approximated by the density multiplied by the area of the region.

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## 7.3 Key Properties of the Joint PDF

Two important properties are stated:

1. Non-negativity

$$f(x, y) \geq 0$$

2. Normalization condition

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

The slides also emphasize an important remark:

> The value of $f(x, y)$ may be greater than 1 because it represents a probability density rather than a probability.
> L15_L17

---

# 8. Worked Example

The lecture presents a detailed example involving a joint PDF.

---

## 8.1 Given

$$f(x, y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Thus the pair $(X, Y)$ takes values in the **unit square**

$$[0, 1]^2$$

L15_L17

---

## 8.2 Task 1: Verify that $f(x, y)$ is a Valid Joint PDF

To verify validity, check normalization.

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

Step 1:

$$\int_0^1 \left( \int_0^1 4xy\, dx \right) dy$$

Step 2:

$$\int_0^1 2y\,dy$$

Step 3:

$$[y^2]_0^1 = 1$$

Since

- $f(x,y) \geq 0$
- the total integral equals 1

the function is a **valid joint PDF**.

L15_L17

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## 8.3 Task 2: Compute $P(A)$

The event is defined as

$$A = \{X < 0.5,\ Y > 0.5\}$$

The probability of this event is

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx$$

Step-by-step evaluation:

$$P(A) = \int_0^{0.5} [2xy^2]_{0.5}^1 dx$$

$$= \int_0^{0.5} 2x(1-0.25)\,dx$$

$$= \int_0^{0.5} \frac{3}{2}x\,dx$$

$$= \frac{3}{16}$$

Thus

$$P(A) = \frac{3}{16}$$

L15_L17

---

## 9. Joint Cumulative Distribution Function (Joint CDF)

Let $X$ and $Y$ be jointly distributed random variables.

The notation

$$X \leq x, \quad Y \leq y$$

represents the event

$$\{X \leq x \text{ and } Y \leq y\}$$

---

## Definition

The **joint cumulative distribution function** is defined as

$$F(x, y) = P(X \leq x, Y \leq y)$$

This function gives the probability that both variables simultaneously take values less than or equal to specified thresholds.

L15_L17

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## 10. Joint CDF – Continuous Case

If $X$ and $Y$ are continuous random variables with joint density $f(x, y)$, then the joint CDF is obtained by integrating the density:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

This integral accumulates the density over the region defined by

$$u \leq x, \quad v \leq y$$

L15_L17

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## Recovering the Joint PDF

Because two variables are involved, the density can be recovered from the CDF using **second-order partial derivatives**:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

L15_L17

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## 11. Joint CDF – Discrete Case

If $X$ and $Y$ are discrete random variables with joint PMF $p(x_i, y_j)$, the joint CDF is computed by summing probabilities:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

The lecture explains the interpretation of this summation:

1. Add all probabilities in the joint PMF table.
2. Include rows where $x_i \leq x$.
3. Include columns where $y_j \leq y$. L15_L17

---

## 12. Properties of the Joint CDF

The joint CDF $F(x, y)$ satisfies several properties.

---

## Property 1: Monotonicity

$$F(x, y)$$

is **non-decreasing**.

If either $x$ or $y$ increases, the value of $F(x, y)$ remains the same or increases. L15_L17

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## Property 2: Lower-Left Boundary

$$F(x, y) = 0$$

at the lower-left corner of the joint range.

If the lower bound is $(-\infty, -\infty)$, then

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

L15_L17

---

## Property 3: Upper-Right Boundary

$$F(x, y) = 1$$

at the upper-right corner of the joint range.

If the upper bound is $(\infty, \infty)$, then

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

L15_L17

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## 13. Summary of Lecture L17

This lecture introduced the fundamental concepts required to analyze **two random variables simultaneously**.

The main concepts developed in the lecture include:

- Motivation for studying joint random variables
- Real-world applications of joint probability models
- Definition of the **joint PMF** for discrete variables
- Representation of joint probabilities using tables
- Example involving **rolling two dice**
- Definition and properties of the **joint PDF** for continuous variables
- Verification and probability computation using a joint density function
- Definition of the **joint CDF**
- Continuous and discrete formulations of the joint CDF
- Important properties of the joint CDF

These concepts provide the mathematical foundation for studying **relationships between random variables**, which is essential for many applications in computing, statistics, and data science.

L15_L17

**Sources**

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